Progress Report: How Foveated Vision Shapes Cognitive Maps in Navigation Agents

Leo Bruneau, Weilun Xu (394449), Rim Abkari, Zyad Aoutir

CS-503 Visual Intelligence — Progress Report (Updated: April 13, 2026)

I. Overview

Research question: Does non-uniform (foveated) visual input reshape the spatial representations learned in a navigation agent's recurrent memory, compared to blind, uniform-resolution, and matched-compute baselines?

Three hypotheses:

H1 Compensatory memory: Foveated agents develop longer-horizon spatial memory to compensate for degraded peripheral vision.

H2 Representational divergence: Different visual inputs produce qualitatively different representational geometries, not just different accuracy levels.

H3 Epistemic gaze: A learned-gaze agent directs fixation toward memory-weak regions, driven by task pressure alone.

Setup: All agents are trained with DD-PPO [1] on PointGoal navigation in Habitat [2], using Gibson-0+ (411 scenes) + MP3D train (61 scenes). All share an identical 3-layer LSTM (512-d hidden) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar). Five conditions vary only the visual input:

- **Blind:** No visual input (replicates Wijmans et al. [3]).
- **Uniform:** Full-resolution 256×256 RGB via ResNet-18.
- **Foveated-fixed:** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff), gaze locked at image centre.
- **Matched-compute:** Low-resolution uniform RGB, total pixel budget exceeding foveated effective budget (controls for information volume).
- **Foveated-learned:** Same blur as foveated-fixed, but gaze position predicted by an MLP from the LSTM hidden state (tests H3).

Architecture note: Wijmans et al. [3] trained only blind agents; they did not train a sighted variant. The DD-PPO sighted agent they reference (from [1]) uses ResNet-50 with depth input. We use ResNet-18 with RGB to keep the model smaller and because our research question requires training multiple visual conditions; the matched-compute baseline controls for total visual information regardless of encoder size.

Analysis: Freeze weights → collect per-step LSTM hidden states → linear probing (Ridge regression, episode-level splits) for spatial variables → cross-condition CKA and probe transfer → behavioral tests (shortcut discovery, memory transplant) → PCA/t-SNE visualization.

II. Training Status

Training targets are set to ensure convergence rather than a fixed frame budget.

Table I
TRAINING STATUS PER CONDITION (AS OF APRIL 12, 2026).

Condition	Frames	Target	Success	SPL	Status
Blind	341M	250M	90.7%	0.550	**Complete**
Uniform	400M	400M	97.6%	0.805	**Complete**
Foveated	155M	250M	98.3%	0.823	Training
Matched	203M	250M	92.9%	0.582	**Complete**
Fov-learned	45M	250M	—	—	Training

Design note: Uniform is trained to 400M (vs. 250M for others) because its ResNet-18 encoder requires more samples to converge. We match convergence level rather than frame count to ensure fair cross-condition comparison.

III. Probing Results (Phase 1 – Baseline)

Full 13-experiment probing analysis has been completed for all four conditions using their best available checkpoints (blind: ckpt.19, uniform: ckpt.33, foveated: ckpt.21, matched: ckpt.30). All probes use full 512-d hidden states (no PCA).

A. Global GPS and Compass Probes (1b)

Table II
GLOBAL LINEAR PROBE ACCURACY (RIDGE, $\alpha = 10$, EPISODE-LEVEL TRAIN/TEST SPLIT). HIGHER R^2 = BETTER SPATIAL ENCODING.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	**0.899**	0.020m	**0.878**	0.8°
Foveated	0.841	0.034m	0.715	2.1°
Matched	0.832	0.033m	0.607	1.3°
Uniform	0.614	0.035m	0.423	1.6°

Key observations:

- Blind dominates baseline probes—GPS/compass are directly in the input, so the LSTM trivially encodes them. Replicates Wijmans et al. [3].
- Among sighted conditions, **foveated ranks highest** on both GPS (0.841) and compass (0.715).
- Uniform is weakest (GPS 0.614)—the full visual stream may act as a crutch, reducing reliance on memory.

B. Distance-to-Goal and Control Tasks (1c, 1e–f)

All selectivity values are well above zero, confirming probes capture genuine spatial structure, not memorization artifacts.

C. Multi-Layer Comparison (1d)

For the blind agent, layer 0 hidden state (h) achieves the highest GPS R^2 (0.958), while layer 1 cell state (c) is a close second (0.962). The top LSTM layer (layer 2, h) matches the default probe target (0.899). Spatial information is distributed across layers but concentrated in layers 0–1.

Table III
DISTANCE-TO-GOAL PROBE AND SELECTIVITY (HEWITT & LIANG CONTROL).

	Dist-to-goal R^2	GPS selectivity	Compass selectivity
Blind	**0.984**	**0.911**	**0.890**
Foveated	0.953	0.880	0.751
Matched	0.977	0.871	0.701
Uniform	0.959	0.763	0.543

IV. PROBING RESULTS (PHASE 2 – H1: COMPENSATORY MEMORY)

A. Path-History Probe (2c)

This probe decodes the agent's position at k steps *in the past* from the current hidden state, testing how far back spatial memory extends.

Table IV
PATH-HISTORY PROBE: R^2 FOR DECODING GPS AT LAG k FROM CURRENT HIDDEN STATE. HIGHER = LONGER WORKING MEMORY.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.536
Foveated	0.841	0.862	0.826	0.737	**0.685**	**0.629**
Matched	0.832	0.841	0.779	0.581	0.475	0.445
Uniform	0.614	0.555	0.393	0.386	0.008	−0.105

This is the strongest finding so far. The foveated agent retains spatial information over longer temporal horizons than any other condition:

- At lag-5, foveated R^2=0.629 vs. blind 0.536, matched 0.445, uniform −0.105.
- Foveated surpasses even blind at lags 4–5, despite blind having GPS directly in its input.
- Uniform memory decays rapidly and becomes uninformative by lag-5.

This is **direct evidence for H1 (compensatory memory)**: degraded peripheral vision forces the foveated agent to build a richer temporal trace of where it has been.

The foveated–matched comparison (0.629 vs. 0.445 at lag-5) isolates the effect of *spatial non-uniformity* from total information volume: it is the uneven quality of the foveated sensor, not simply having less visual information, that drives compensatory memory.

B. Visited-Region Probe (2d)

Table V
VISITED-REGION PROBE: CAN THE HIDDEN STATE PREDICT WHICH SPATIAL CELLS THE AGENT HAS VISITED?

	R^2	Binary accuracy
Blind	**0.834**	99.8%
Uniform	0.826	99.8%
Matched	0.808	99.8%
Foveated	0.801	99.8%

All conditions track visited regions near-perfectly (>99.8% binary accuracy). R^2 values are similar, suggesting this is a basic capability shared by all agents regardless of visual input.

C. Occupancy Map Reconstruction (2e)

Table VI
OCCUPANCY DECODING FROM HIDDEN STATES (20×20 GRID).

	R^2	Binary accuracy
Blind	−0.144	59.6%
Matched	−0.379	58.1%
Foveated	−0.397	55.2%
Uniform	−0.300	56.3%

Occupancy decoding is weak across all conditions (all $R^2 < 0$). None of the agents build detailed layout maps in their hidden states. This may require more episodes, a finer grid, or a non-linear decoder to detect.

D. Place Cells (5a)

Table VII
PLACE-CELL-LIKE UNITS (SPATIAL INFORMATION > 1 BIT) OUT OF 512 HIDDEN UNITS.

	Place cells (>1 bit)
Foveated	**475**/512
Blind	473/512
Uniform	473/512
Matched	461/512

Nearly all units carry spatial information across all conditions. Matched has slightly fewer place-cell-like units (461 vs. ∼473).

V. PROBING RESULTS (PHASE 3 – H2: REPRESENTATIONAL DIVERGENCE)

A. Cross-Condition CKA (3a)

Linear CKA measures representational similarity between conditions (1.0 = identical geometry, 0.0 = unrelated). We compute CKA on the top LSTM hidden states (h_layer_2) using matched episodes across all condition pairs.

Table VIII
LINEAR CKA SIMILARITY (TOP-LAYER HIDDEN STATE, $n \approx 1{,}910$ STEPS).

	Blind	Uniform	Foveated	Matched
Blind	—	0.0023	0.0023	0.0011
Uniform		—	**0.0007**	0.0017
Foveated			—	0.0011

Key observations:

- All CKA values are near zero (<0.003), indicating that each condition develops a **fundamentally different representational geometry** for encoding space.
- Blind is the most dissimilar to all sighted conditions—consistent with its reliance on proprioceptive rather than visual features.
- The closest pair is uniform–foveated (0.0007), yet still near zero, showing that even sighted agents with different visual sampling diverge strongly.

B. Probe Transfer (3b)

We train a GPS probe on one condition's hidden states and evaluate it on another's. Self-accuracy (∼0.96–0.97) confirms probes work; cross-condition transfer tests whether spatial information is encoded in a shared format.

Table IX
PROBE-TRANSFER R^2: GPS PROBE TRAINED ON ROW, TESTED ON COLUMN.

Train \ Test	Blind	Uniform	Foveated	Matched
Blind	**0.973**	−9.72	−2.71	−15.72
Uniform	−7.26	**0.973**	−13.53	−11.47
Foveated	−11.25	−5.12	**0.963**	−10.66
Matched	−5.70	−11.74	−15.52	**0.959**

Cross-condition transfer is **catastrophically negative** in all cases, confirming that spatial information is not encoded in a shared linear subspace across conditions. Combined with near-zero CKA, this provides **strong evidence for H2**: visual input type fundamentally shapes the representational geometry used to encode space, not just the amount of spatial information stored.

VI. Behavioral Test: Shortcut / Cognitive Map

The shortcut discovery test [4] evaluates whether an agent's accumulated spatial experience *functionally* aids navigation—the behavioral signature of a cognitive map. For each scene, we run 10 episodes sequentially in two conditions:

- **Reset** (standard): LSTM hidden state zeroed between episodes.
- **Persistent** (cognitive-map): hidden state carries over, allowing spatial knowledge to accumulate across episodes within the same scene.

If persistent > reset, the agent's recurrent representations encode reusable spatial knowledge. Results across conditions (20 Gibson scenes, 10 episodes/scene each):

Table X
SHORTCUT / COGNITIVE-MAP BEHAVIORAL TEST (20 SCENES × 10 EPISODES PER CONDITION). Δ = PERSISTENT − RESET.

	Reset (standard)		Persistent		Δ Success
Condition	Success	SPL	Success	SPL	
Blind	0.78	0.432	0.24	0.145	**−0.53**
Matched	0.86	0.511	0.75	0.421	−0.12
Uniform	**0.95**	**0.786**	**0.93**	**0.592**	−0.02
Foveated	*Pending (resubmitted after grid-buffer fix)*				

Key findings:

1) **Persistent memory hurts all conditions**, but the severity varies by an order of magnitude: blind −53%, matched −12%, uniform −2%.
2) **Vision acts as a grounding signal.** Sighted agents—especially uniform—maintain near-perfect success even with out-of-distribution hidden states. The visual stream provides real-time spatial evidence that overrides stale memory, preventing the catastrophic failure seen in blind.
3) **SPL drops even when success holds.** Uniform loses −0.194 SPL despite only −2% success loss, indicating that accumulated state increases path inefficiency (more wandering) without preventing goal-reaching.
4) **Map-building is negative for all conditions.** Within the persistent condition, later episodes are *worse* than earlier ones (blind: −0.146, matched: −0.127, uniform: −0.112), confirming progressive state corruption rather than spatial learning.

Interpretation: No condition builds a reusable cognitive map across episodes. The blind agent's LSTM representations are *episode-scoped path-integration states*: "where I am relative to where I started this episode." Carrying this across episode boundaries places the LSTM in an out-of-distribution regime, collapsing performance. Sighted agents degrade more gracefully because visual input re-anchors the hidden state each step, but even they do not *benefit* from persistent memory.

VII. Summary of Hypotheses

Table XI
HYPOTHESIS STATUS BASED ON CURRENT RESULTS.

Hypothesis	Status	Key evidence
H1: Compensatory memory	**Supported**	Foveated lag-5 R^2=0.629 (best across all conditions)
H2: Representational divergence	**Supported**	All CKA <0.003; cross-transfer catastrophically negative
H3: Epistemic gaze	In progress	Learned-gaze training at 15M/250M
Additional finding: Cognitive map (behavioral)	**Negative (all)**	Persistent LSTM hurts all agents; blind −53%, uniform −2%. Vision grounds but doesn't build m

VIII. Next Steps

1) **Complete training** (in progress): Foveated (155M→250M), fov-learned (45M→250M). Blind, uniform, and matched complete.
2) **Foveated shortcut eval** (queued): Resubmitted after grid-buffer fix. Will complete the cross-condition behavioral comparison.
3) **Foveated-learned training** (in progress, 15M/250M): Config, policy, and slow-gaze approximation implemented. Running on cs-503 QOS (∼5 days to target).
4) **Re-probe after convergence:** Re-run full 13-experiment probing suite on final checkpoints for definitive comparison. Current probing results use mid-training checkpoints.
5) **Visualization:** PCA/t-SNE of hidden states colored by position and condition. Script ready (`scripts/probing/visualize.py`).
6) **Memory transplant test:** Cross-condition hidden state transplant (`scripts/eval/transplant.py`) to test whether spatial representations are interchangeable across conditions (complements CKA/transfer probing with a behavioral test).

IX. Infrastructure and Tooling

- **Cluster:** SCITAS Izar (EPFL), V100 GPUs. cs-503 QOS: max 2 submitted, max 1 running, 7d12h wall-time.
- **Probing suite:** 13 experiments per condition (GPS, compass, distance-to-goal, multi-layer, selectivity, path-history lags 0–5, visited-region, occupancy map, place-cell rate maps). All implemented in `scripts/probing/`.
- **Cross-condition analysis:** Linear CKA + probe transfer (`scripts/probing/analyze_cross.py`).
- **Behavioral tests:** Shortcut discovery with persistent vs. reset LSTM (`scripts/eval/shortcut.py`).
- **Visualization:** PCA/t-SNE + path-history decay plots (`scripts/probing/visualize.py`).
- **Key lesson:** PCA-32 dimensionality reduction severely underestimates probe accuracy (blind GPS R^2 drops from 0.899 to 0.580). All final probes use full 512-d hidden states.

X. Progress Log

Weeks 1–2 (Mar 24–Apr 6): Codebase setup; implemented all 5 visual conditions; launched training for blind, uniform, foveated-fixed, matched-compute on Izar.

Week 3 (Apr 7–12): Probing infrastructure built (13-experiment suite). Discovered PCA-32 artifact, switched to full 512-d. Completed Phase 1 probing for all 4 training conditions. Ran cross-condition CKA (H2 supported). Fixed eval pipeline bugs (stale code path, checkpoint key format). Completed blind shortcut eval—persistent LSTM memory catastrophically hurts (-53% success), showing episode-scoped path integration rather than reusable cognitive maps. Launched foveated-learned training (H3). Submitted shortcut eval for 3 sighted conditions.

Week 4 (Apr 13–): *Planned:* Complete remaining training (foveated, uniform, fov-learned). Collect sighted shortcut results. Re-probe with final checkpoints. Generate visualizations. Run memory transplant. Begin final paper.

References

[1] E. Wijmans, A. Kadian, A. Morcos, S. Lee, I. Essa, D. Parikh, D. Batra, and M. Savva, "DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames," in *International Conference on Learning Representations (ICLR)*, 2020.

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[3] E. Wijmans, M. Savva, I. Essa, S. Lee, A. Morcos, and D. Batra, "Emergence of maps in the memories of blind navigation agents," in *International Conference on Learning Representations (ICLR)*, 2023.

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